

The Rotating Shallow Water Equations

The shallow water equations provide a nice, relatively simple analog to the more complete primitive and Boussinesq systems. They might be used to describe a layer of constant density fluid in which the horizontal scales of interest are large compared to the layer thickness. Shallow water models rewritten using pressure as a vertical coordinate can also be used as a simple model of the troposphere. Linearized versions of the shallow water equations also describe the dynamics of a baroclinic mode (e.g., as might be obtained from a separation of variables) of the primitive equations. Finally, the shallow water equations also come up in modelling things as diverse as volcanoes and accretion disks, and a vertical stack of shallow water layers is sometimes used as an analog to a stratified fluid. In the case of a two-layer stack with one thick and one thin layer, the motion of the thinner layer is well-described by the shallow water equations, but with a “reduced gravity” (e.g., with gravity, g , replaced by $g' = g\Delta\rho/\rho$ where ρ is density and $\Delta\rho$ is the density jump between the two immiscible layers).

An outline of this section is as follows:

1. Underlying assumptions and a quick derivation
2. The case of zero f
3. Potential vorticity
4. Linear wave and geostrophic modes
5. The Rossby (or geostrophic) adjustment problem

Assumptions and derivation: Because it is easy to visualize, we will take the physical system under consideration to be a layer of constant density water. What is meant by ‘shallow’ is that the layer thickness is small compared to the horizontal length scales under consideration. Formally, what is required is that $H^2/L^2 \ll 1$, where H is the layer thickness and L is a relevant horizontal length scale (the width of the Gulf Stream, say). It is this thin vertical-to-horizontal aspect ratio that allows us to make the hydrostatic approximation. The hydrostatic approximation gives that $p_z = \rho g$ where density, ρ , is assumed constant, g is the acceleration of gravity, p is pressure, and the subscript denotes partial differentiation.

For now, we assume a flat bottom at $z = -H$ and take $z = \eta(x, y, t)$ to be the vertical position of the upper interface. The layer thickness is then $d = H + \eta$. For many applications, η/H is small and this can be used to simplify the equations, but this need not always be the case. Above $z = \eta$, $p = p_{\text{atm}}$, which we will take to be constant and without loss of generality, we take $p_{\text{atm}} = 0$. Then, on a horizontal surface below $z = \eta$ ($z = z_0 < 0$, say) we have that $p = \rho g(\eta - z_0)$. In words, pressure is given by the weight of the water column. The horizontal pressure gradient force, $-(1/\rho)\nabla_{\text{h}}p$ then works out to be $-g\nabla\eta$.

In other words, the HPGF is independent of depth and is proportional to the slope of the interface height field, η . We also assume the horizontal velocity to be depth independent, i.e., $\mathbf{u}_z = 0$. The idea this is that if $\mathbf{u} = u\hat{\mathbf{x}} + v\hat{\mathbf{y}}$ is initially independent of z , and all forces acting on it are also z -independent, then $\mathbf{u}$ should remain independent of z .

We also make the (3-d) incompressibility assumption: $\nabla \cdot \mathbf{u}_{3d} = 0$, where $\mathbf{u}_{3d} = u\hat{\mathbf{x}} + v\hat{\mathbf{y}} + w\hat{\mathbf{z}}$. We will denote the 2-d part of this by $\mathbf{u}$. Taking a z derivative gives

$$\partial_z(\nabla \cdot \mathbf{u}_{3d}) = \nabla \cdot \mathbf{u}_z + w_{zz} = 0.$$

Thus, if $\mathbf{u}_z$ is assumed to vanish, then so must w_{zz} . This implies $w_z = (w_{\text{top}} - w_{\text{bot}})/d$. Since we have assumed a flat bottom, the no normal flow condition gives that $w_{\text{bot}} = 0$. The definition of η as the z -position of the top, together with the definition of the Lagrangian derivative give that $w_{\text{top}} = D_t \eta$. In other words, η can be thought of as the z -position of a material parcel which happens to be at the top of the water column, and the Lagrangian derivative of the position vector of a material parcel is the velocity.

We thus have that

$$\frac{D}{Dt} \eta + d(\nabla \cdot \mathbf{u}) = 0,$$

which is often called the continuity or mass continuity equation. The full system is this, combined with the horizontal momentum equation.

$$\begin{aligned} \frac{D}{Dt} \eta + (H + \eta) \nabla \cdot \mathbf{u} &= 0 \\ \frac{D}{Dt} \mathbf{u} + f \hat{\mathbf{z}} \times \mathbf{u} &= -g \nabla \eta. \end{aligned} \tag{1}$$

Above, the Lagrangian derivative is given by $D_t = \partial_t + \mathbf{u} \cdot \nabla$, f is the Coriolis parameter, and we have made the traditional approximation (a common simplification to the Coriolis force). A few comments seem appropriate

1. Since $D_t H = 0$, we can rewrite the mass equation as $D_t d + d \nabla \cdot \mathbf{u}$. This is equivalent to a 2-d version of the full mass continuity equation (e.g., without making the incompressibility approximation). That is, except for the operators being in two instead of three dimensions, the shallow water mass continuity equation looks like $D_t \rho + \rho \nabla \cdot \mathbf{u}_{3d} = 0$. If this Lagrangian form is not familiar, the two equations can be rewritten in Eulerian form: $\eta_t + \nabla \cdot (d \mathbf{u}) = 0$ and $\rho_t + \nabla \cdot (\rho \mathbf{u}_{3d}) = 0$.
2. Adding bathymetry is straightforward; simply, $d = H + \eta$ is replaced by $d = H + \eta - h_b(x, y)$, where $z = -H + h_b$ gives the bathymetry. It is worth going through the algebra to show this if it does not seem obvious.
3. The momentum equation can also be written in vorticity-Bernoulli form. Vorticity is the curl of velocity, and its vertical component is given by $\zeta = v_x - u_y$. The Bernoulli function is $B = g\eta + \mathbf{u} \cdot \mathbf{u}/2$. In vorticity-Bernoulli form, the shallow water momentum equation becomes

$$\mathbf{u}_t + (f + \zeta) \hat{\mathbf{z}} \times \mathbf{u} = -\nabla B.$$

To show this, one can either look for useful vector identities or write out the equations in component form, and perform some simple manipulations. Taking the latter approach, the $\hat{\mathbf{x}}$ -component of the momentum equation is

$$u_t + uu_x + vu_y - fv = -g\eta_x.$$

We then add and subtract vv_x to get

$$u_t + (uu_x + vv_x) - v(v_x - u_y) - fv = -g\eta_x,$$

and a simple rearrangement gives

$$u_t - (f + \zeta)v = -B_x.$$

In vector notation, the shallow water momentum equation can thus be written as

$$\mathbf{u}_t + (\mathbf{u} \cdot \nabla) \mathbf{u} + f \hat{\mathbf{z}} \times \mathbf{u} = -g \nabla \eta$$

or,

$$\mathbf{u}_t + (f + \zeta) \hat{\mathbf{z}} \times \mathbf{u} = -\nabla B.$$

Chapter 4: The nonrotating limit

In the nonrotating limit $f \rightarrow 0$, the momentum equation reduces to

$$\mathbf{u}_t + \mathbf{u} \cdot \nabla \mathbf{u} = -g \nabla \eta$$

which represents, so to speak, shallow-water dynamics in a bathtub. At any rate, it may be a good deal more intuitive as a starting point, and we shall use it to explore the notion of a polarisation relation. First, let us specialise to the linear 1D case, where we have the system:

$$u_t = -g \eta_x$$

$$\eta_t = -H u_x$$

which can be combined together to yield, as we have seen, “the” wave equation:

$$\eta_{tt} = -g H \eta_{xx}$$

$$u_{tt} = -g H u_{xx}$$

where $c = \pm \sqrt{gH}$. Now the point here is that we can take a real Fourier transform and, in theory, propagate the waves; but for each k we have *two* values of $\omega = \pm \sqrt{gH}$.—an equality known as the *dispersion relation*. These are not redundant like the positive and negative values of k , because if k and ω are of the same sign the wave propagates rightward, but otherwise it goes left. But note also that we have physical fields and thus two Fourier coefficients, $\hat{u}_k$ and $\hat{\eta}_k$. (If we wish, we could write these as 4 coefficients, for the sine and cosine parts of each field). We now introduce a simple procedure, in fact a change of basis, where we go from two coefficients for the two fields towards two coefficients for the two waves.

The derivation here is quick enough. First we undertake a real Fourier transform of the equations, and for a particular k and one of its two ω s:

$$-i\omega \hat{u} e^{i(kx - \omega t)} = -igk \hat{\eta} e^{i(kx - \omega t)}$$

Weeding out the common factors we arrive at

$$\hat{u} = g \frac{k}{\omega} \hat{\eta} = \frac{g}{c} \hat{\eta}$$

This is our *polarisation relation*.

But remember that c takes two different values, $\pm \sqrt{gH}$. Associating this with two waves, each with their own amplitude and phase, we write:

$$\hat{u}_k = \hat{u}_k^+ + \hat{u}_k^-$$

$$\hat{\eta}_k = \hat{\eta}_k^+ + \hat{\eta}_k^-$$

And substituting our polarisation relation in here, using $\omega = \sqrt{gH}$ for the right-propagating waves and $\omega = -\sqrt{gH}$ for the left-propagating ones:

$$\hat{u}_k = \sqrt{\frac{g}{H}} (\hat{\eta}_k^+ - \hat{\eta}_k^-)$$

$$\hat{\eta}_k = \hat{\eta}_k^+ + \hat{\eta}_k^-$$

which is a system of two linear equations in two variables. With all this under your belt, go try out Exercise 4!

Chapter 5: Potential vorticity Vorticity is the curl of velocity. In general, given the vorticity, divergence, and boundary conditions, the total flow is known. Since when making the incompressibility approximation in 3-d, there is no divergence, then the vorticity alone describes the flow. Flow in the atmosphere and ocean tends to be quasi two dimensional and, for many applications, the horizontal divergence can be considered small compared to the vertical vorticity (i.e., to the vorticity associated with the horizontal flow). In this limit (essentially), potential vorticity plays a special role in the dynamics: knowing the potential vorticity, and boundary conditions, allows one to infer the velocity field. Conversely, knowing the velocity field tells us how the potential vorticity field evolves in time.

We start by taking the curl of the momentum equation to form the “vorticity equation”. Since the motion is essentially two-dimensional (w is small compared to u and v), we restrict attention to the vertical component of vorticity, $\zeta = v_x - u_y$. Taking the curl of the shallow water momentum equation then amounts to subtracting the y derivative of the u -momentum equation from the x derivative of the v -momentum equation. That is,

$$\begin{aligned} [v_t + (f + \zeta)u] &= -B_y]_x \\ -[u_t - (f + \zeta)v] &= -B_x]_y \end{aligned} \quad (2)$$

$$\zeta_t + \mathbf{u} \cdot \nabla \zeta = -(f + \zeta) \nabla \cdot \mathbf{u}.$$

Above, I have taken f to be constant. More generally, one gets

$$D_t(f + \zeta) = -(f + \zeta) \nabla \cdot \mathbf{u}.$$

Note that $\nabla \cdot \mathbf{u}$ appears both in the vorticity equation and in the mass equation. Eliminating it gives us the potential vorticity equation. For example, in the vorticity equation above, we substitute $(-1/d)D_t d$ for $\nabla \cdot \mathbf{u}$ to find

$$\begin{aligned} \frac{1}{f + \zeta} \frac{D}{Dt} (f + \zeta) - \frac{1}{d} \frac{D}{Dt} (d) &= 0 \\ \frac{D}{Dt} \ln(f + \zeta) - \frac{D}{Dt} \ln d &= 0 \\ \frac{D}{Dt} \ln[(f + \zeta)/d] &= 0 \\ \frac{D}{Dt} [(f + \zeta)/d] &= 0. \end{aligned} \quad (3)$$

The scalar quantity, $(f + \zeta)/d$ is called potential vorticity; we shall denote it Q . It is conserved in the Lagrangian sense, meaning it is conserved following a column of fluid as it is advected

around by the flow. (Note that in the shallow water framework, the concept of a parcel gets replaced with that of a “fluid column”, so we say that potential vorticity is conserved “following a fluid column”.) Conservation of volume gives that the volume of a column is conserved.

Repeating this exercise for a linearized version of the equations gives

$$\begin{aligned} [v_t + fu &= -g\eta_y]_x \\ -[u_t - fv &= -g\eta_x]_y \\ \hline \zeta_t &= -f\nabla \cdot \mathbf{u}. \end{aligned} \tag{4}$$

Then, for the flat bottom case, $\nabla \cdot \mathbf{u} = -\eta_t/H$ and we conclude that

$$(\zeta - \frac{f\eta}{H})_t = 0.$$

The quantity in the parenthesis is conserved, and is a linear analogue to Q . In fact, if we linearise Q we get something proportional to this. Taking both $O(\zeta/f)$ and $O(\eta/H)$ to be small:

$$\begin{aligned} \frac{f + \zeta}{H + \eta} &\approx \frac{f + \zeta}{H} (1 - \frac{\eta}{H}) \\ &\approx \frac{1}{H} (f + \zeta - f\eta/H) + \text{small}. \end{aligned} \tag{5}$$

Ignoring the large constant term (f) and division by the constant, H , we see that $\zeta - f\eta/H$ is equivalent to potential vorticity under these conditions. We shall denote this linearised term q , and refer to it as the potential vorticity. This result ignores bathymetry and assumes H to be large compared to η . Thus in this linear limit, potential vorticity is set by initial conditions—and it never changes. More realistically, it doesn’t change appreciably over some short time scale of interest. Essentially, it doesn’t change until enough time has passed for advection (the nonlinear term that we have unceremoniously jettisoned) to have moved it around enough such that its distribution is substantially altered.

Taking A to be a cross-sectional area, volume is given by Ad . That is, $A \propto 1/d$. Physically, then, we can think of potential vorticity as $(f + \zeta)A$. Recall that $f + \zeta$ corresponds to twice the angular velocity of the local spin about the vertical axis. Then, since the moment of inertia for a cylinder goes like the square of its radius, we can interpret conservation of potential vorticity as each parcel (column) conserving its angular momentum (as measured in an inertial reference frame).

Obviously, if we were to include forcing and dissipation terms in the momentum and mass equations, we would find forcing and dissipation terms in the potential vorticity equation as well. Two comments are important to make here. One is that the conservation of potential vorticity we derived above has analogies in the more complete system. That is, the primitive equations, Boussinesq equations and in fact even the fully compressible equations all have potential vorticity conservation laws. Essentially, it is only required that viscous and diffusive effects are ignored and that the equation of state is “sufficiently nice”. Examples of nice equations of state include the dry ideal gas law and linearized versions of the equation of state of sea water. A second comment is that the distribution of potential vorticity (i.e., the function $Q(x, y)$ at some time, t), together with boundary conditions gives a strong constraint on the “balanced” or nearly-geostrophic part of the total flow. We shall make the relationship between potential vorticity and geostrophic clearer in what follows.

Exercise 5: Purely pen-and-paper:

1. Explain your observations of the vorticity field in the nonrotating case (exercise 4).
2. Consider quasi-zonal flow over a meridionally oriented ridge (e.g., $h = h(x)$). Take the flow to be approximately geostrophic and time independent. If, as it flows over the ridge, a fluid column conserves its potential vorticity, then layer thickness, d , gets smaller (or “squashes”) as the current flows upslope and then stretches again as it goes back down. Make a sketch of the flow under the following assumptions: i) this occurs on a Northern Hemisphere β -plane and changes in d are compensated mainly by changes in f (i.e., so as to keep the ratio $(f + \zeta)/d$ constant and ii) it is changes in ζ (rather than f) that compensate for changes in d as the flow goes over the ridge.
3. For the Boussinesq equations, potential vorticity is given by $(f + \zeta)N^2 + (f\hat{z} + \boldsymbol{\omega}) \cdot \nabla b$, where $\boldsymbol{\omega}$ is the three dimensional relative vorticity: $\boldsymbol{\omega} = (w_y - v_z)\hat{x} + (u_z - w_x)\hat{y} + \zeta\hat{z}$ and N^2 is the buoyancy frequency (which you can take as constant). Speculate as to how this might simplify for the primitive equations, i.e., for the case $H^2/L^2 \ll 1$. [Note: this problem can be skipped if you are not familiar with the Boussinesq and hydrostatic approximations.]
4. Consider a 2-layer stack of shallow water layers. Let us denote the top and interior interfaces by η^{surf} and $\eta^{interface}$, respectively. Show that the horizontal pressure gradient force in the lower layer is given by $-g\nabla\eta^s - g'\nabla\eta^i$, where $g' \equiv (\rho_2 - \rho_1)/\rho_0$ is called the reduced gravity and the superscripts s and i refer to surface and interface. Note: it is assumed that $\rho_2 - \rho_1 \ll \rho_0$, i.e., that the density difference between the two layers is small compared to the reference density. Further, try to write out the complete system for this 2-layer shallow water system. (There are six scalar variables, i.e., the two interfaces and the two velocity components in each layer, so a total of six scalar equations are needed.) To simplify, assume the bottom to be flat.

Chapter 6: f-plane theory:

Poincaré waves The Poincaré wave is the generalisation of the nonrotating wave to the f -plane. We start with the linear shallow water system on an f -plane (i.e., with f taken to be constant),

$$\begin{aligned} u_t - fv &= -g\eta_x \\ v_t + fu &= -g\eta_y \\ \eta_t + H(u_x + v_y) &= 0. \end{aligned} \tag{6}$$

Remember that we had previously discussed inertial oscillations, where we suppressed the pressure term. In that case, the net effect of f was to turn u into v and vice-versa. We have also discussed the nonrotating case, where we suppressed f instead, getting a different limit. Poincaré waves respond to both these mechanisms, as we shall see.

Above, we took the curl of the momentum equation and substituted from the mass equation to get that $\partial_t(\zeta - f\eta/H) = 0$. Here, we instead take the divergence of the momentum equation to get

$$(u_x + v_y)_t - f\zeta = -g\nabla^2\eta.$$

Substituting from the mass equation leads to $\eta_{tt} - gH\nabla^2\eta + fH\zeta = 0$. The first two terms have obvious analogues in the non-rotating shallow water wave equation. To get a single equation

in the variable, η , we use conservation of potential vorticity, which gives that $\zeta_t = f\eta_t/H$, or $\zeta = f\eta/H + C(x, y)$, where $C(x, y)$ is given by the initial potential vorticity distribution. Finally, let us assume we are interested in waves that have propagated in from somewhere else, and that prior to their arrival, the fluid in our region of interest was at rest. This sets $C \rightarrow 0$ (a *crucial* step), thus giving the Poincaré Wave Equation:

$$\eta_{tt} - c^2 \nabla^2 \eta + f^2 \eta = 0.$$

Assuming solutions of the form $\eta = Ae^{i(kx+ly-\omega t)}$ then gives the dispersion relation:

$$\omega^2 = c^2(k^2 + l^2) + f^2.$$

A few comments:

1. the frequency is bounded below by f .
2. a natural length scale, $L_d = c/f$, appears. This is called the Rossby radius of deformation.
3. long waves ($k^2 + l^2 \ll L_d^{-2}$) have frequencies close to f .
4. short waves ($k^2 + l^2 \gg L_d^{-2}$) behave similarly to non-rotating gravity waves. Verify this limit from the PDE, too.
5. in the 1-d case $\omega^2 = c^2 k^2 + f^2$.

Polarisation relations: Here, we want to find the u and v that ‘go with’ our solutions described above for η . This is most straightforwardly done in 1-d, so let us take $\partial_y \rightarrow 0$; equivalently, $l = 0$. Note that this does not imply that we can take $v = 0$; it implies that v (and the other fields) varies with x but not y . Our equations become

$$\begin{aligned} u_t - fv &= -g\eta_x \\ v_t + fu &= 0 \\ \eta_t + Hu_x &= 0. \end{aligned} \tag{7}$$

Recalling we had assumed $\eta = Ae^{i(kx-\omega t)}$ (where A is a complex constant and ‘real part of’ is implicit). The mass equation then gives $u = (\omega/Hk)Ae^{i(kx-\omega t)}$. In other words, η and u are in phase, with their amplitudes U_0, η_0 (e.g., $\eta_0 = |A|$) given by

$$U_0 = \frac{\omega}{Hk} \eta_0.$$

Similarly the v momentum equation gives that $i\omega V_0 = fU_0$, or

$$V_0 = \frac{-if}{Hk} \eta_0.$$

For example, if $u = U_R \cos(kx - \omega t) - U_I \sin(kx - \omega t)$, then $v = (f/\omega)[U_I \cos(kx - \omega t) + U_R \sin(kx - \omega t)]$.

For short waves, this reduces to $U_0/c \approx \eta_0/H$ with V_0 small (i.e., it reduces to the nonrotating result, as we expect). For long waves, we find a divide-by-zero problem as $k \rightarrow 0$. If we take $k = 0$, then our momentum equations become $u_t = fv$; $v_t = -fu$, corresponding to inertial oscillations in which the pressure gradient term is absent (and that η remains zero is confirmed by

the mass equation). For waves that are long with respect to L_d , but with k finite, we conclude that these “near-inertial oscillations (or waves)” have much more robust signatures in velocity than they do in pressure (or η).

The Rossby Adjustment problem: This is also called the geostrophic adjustment problem. Note that the f -plane problem couples all three variables together, so we should expect *three* wave modes for each k , not just the two Poincaré modes discussed above. In fact, it turns out that the third mode has $\omega = 0$ and it shows up as a stationary flow in the famous *geostrophic balance*. This mode did not show up in the foregoing treatment because of the crucial assumption that the initial potential vorticity distribution was everywhere zero. We shall now relax that assumption: the core question will be how to devise a polarisation-relation type argument to isolate the geostrophic mode (also called the Rossby mode) from the Poincaré modes.

We consider an initial value problem, for example with $\eta = \eta_I(x, y)$ and $u, v = 0$ at $t = 0$. The classic problem is done in 1-dimension called the ‘dam break problem’, e.g., with $\eta_I = +\eta_0$ for $x > 0$ and $\eta_I = -\eta_0$ for $x < 0$. From our discussion above, we know that q_I is given by $-f\eta_I/H$ at $t = 0$ and does not change as time progresses. This implies that, at later times, $\zeta - f\eta/H = \mp f\eta_0/H$, with the sign being negative for $x > 0$ and positive for $x < 0$. Notice that this equation has no time dependence. That is, it describes a steady state that we might expect at least in some neighborhood around $x = 0$ if any transients eventually propagate away from our area of interest. If we recall our (linear) equations of motion, this steady solution must obey

$$\begin{aligned} -fv &= -g\eta_x \\ +fu &= 0 \\ +Hu_x &= 0. \end{aligned} \tag{8}$$

In other words the flow is geostrophic and nondivergent. From this, it is easy to see that $\zeta = (g/f)\eta_{xx}$, so that our steady state equation is given by

$$\bar{\eta}_{xx} - L_d^{-2}\bar{\eta} = -L_d^{-2}\bar{\eta}_I,$$

where the overbar indicates ‘steady’ (or time-averaged). Notice what we have done here. In solving for $\bar{\eta}$ we have found the part of η_I which sits there forever. Using geostrophic balance, we can calculate the similarly steady velocities $(\bar{u}, \bar{v})$. The rest can now be split up into $\hat{\eta}^+$ and $\hat{\eta}^-$, etc., for each k , and we can calculate the leftward and rightward Poincaré waves as we did before.

The solution can be thought of as having homogeneous and particular parts, with the latter trivially corresponding to $\bar{\eta}_p = \eta_I$. The homogeneous part of the solution can also be found (essentially by inspection) to be $\bar{\eta}_h = Ae^{x/L_d} + Be^{-x/L_d}$. We choose A and B by making the total solution well behaved at $x = 0$ and at $x = \pm\infty$. This leads to

$$\bar{\eta} = \begin{cases} +\eta_0(1 - e^{-x/L_d}), & x > 0 \\ -\eta_0(1 + e^{+x/L_d}), & x < 0. \end{cases}$$

The difference $\eta_I - \bar{\eta}$ gives initial conditions for the transients. Note that initial conditions for u, v also must be respected. Above, we had declared these to be zero. It is straightforward to verify that $\bar{u}$ is zero and $\bar{v} = (g/f)\bar{\eta}_x$. Initial conditions for the transients are then

$$\bar{\eta}_{\text{Transient}} = \begin{cases} +\eta_0 e^{-x/L_d}, & x > 0 \\ -\eta_0 e^{+x/L_d}, & x < 0, \end{cases}$$

$$\bar{v}_{\text{Transient}} = \begin{cases} \frac{g\eta_0}{fL_d} e^{-x/L_d}, & x > 0 \\ \frac{-g\eta_0}{fL_d} e^{x/L_d}, & x < 0, \end{cases}$$

with $u_{\text{Transient}} = 0$.

To finish the problem, we need to project initial conditions on to Poincaré wave modes. For this, of course, we will have to recast initial conditions so as to be periodic, e.g., with step function above becoming a square wave of length L . There are then an infinite set of dams breaking at $t = 0$. For large L , however, we expect to recover the same behaviour as in the infinite domain, at least for a while. Eventually, transients from one dam break will propagate into our ‘region of interest’ near the dam break at $x = 0$.

The usefulness of the periodic extension is that we can project initial them onto sinusoids, i.e., onto the Fourier modes. Let us consider the projection on to a particular wavenumber, k_0 . From symmetry properties, we deduce that $\eta_{\text{Transient}}$ projects onto sines but not cosines. But even for a sine wave with $k = k_0$, we have two independent solutions: $\eta = A \sin(k_0 x - \omega t) + B \sin(k_0 x + \omega t)$. Using the polarization relation between η and u that we found above, we get that at $u(t = 0) = 0$ provided $A = B$. In other words the projection is equal on to the left- and right-propagating modes. It is left as an exercise to show that the two signals in v are additive. We conclude, then, that $A = B$, with both corresponding to half of the amplitude found from projecting initial conditions in $\eta_{\text{Transient}}$ on to wavenumber k_0 .

Exercises: You can proceed to exercises 6.1 through to 6.3 at this juncture.

The relevance of the geostrophic adjustment problem to meteorology has to do with how observations are incorporated into weather forecast models. For example, Imagine we had a ‘true’ simulation of the shallow water equations in some particular configuration (maybe including forcing and dissipations terms) and a twin model in which, say, initial conditions or forcing functions were imperfectly known. Even if initial conditions for the twin were close the ‘truth’ initially, after a while we expect the two systems to diverge (in a manner similar to weather forecasts diverging from the weather). To correct for this, say we are allowed to use observations of η (but not of u, v) from the truth simulation to correct η in the twin simulation. Then, if the u and v that ‘go with’ the η field that we are assimilating into the twin model don’t match the u and v of the truth model, the assimilation could make twin diverge further from truth. In particular, if u and v in truth happened to be in geostrophic balance, then we will have essentially added Rossby adjustment type initial conditions superposed on whatever else is going on in the twin simulation.

A matrix approach:

It is perhaps more useful to recast this as an eigenvalue problem. Rewriting our equations using $\phi = g\eta$ and $c^2 = gH$ gives

$$\begin{aligned} \partial_t u - f v &= -\phi_x \\ \partial_t v + f u &= -\phi_y \\ \phi_t + c^2(u_x + v_y) &= 0. \end{aligned} \tag{9}$$

Next, taking $u = U_0 e^{i(kx - \omega t)}$, $v = V_0 e^{i(kx - \omega t)}$, $\eta = \Phi_0 e^{i(kx - \omega t)}$, and substituting back gives the algebraic equation,

$$\begin{pmatrix} -i\omega & -f & ik \\ f & -i\omega & il \\ ic^2k & ic^2l & -i\omega \end{pmatrix} \begin{pmatrix} U \\ V \\ \Phi \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \quad (10)$$

which can also be written as

$$\begin{pmatrix} 0 & if & ck \\ -if & 0 & cl \\ ck & cl & 0 \end{pmatrix} \begin{pmatrix} U_0 \\ V_0 \\ \frac{1}{c}\Phi_0 \end{pmatrix} = \omega \begin{pmatrix} U_0 \\ V_0 \\ \frac{1}{c}\Phi_0 \end{pmatrix}, \quad (11)$$

which is easily recognized as an Eigenvalue problem. The change of variables from Φ to Φ/c makes it so that terms in the three equations have the same units. The Eigenvalues are the ω 's that 'work' and the Eigenvectors give the polarization relations. To find the Eigenvalues, we use the solvability condition,

$$\begin{vmatrix} -i\omega & -f & ick \\ f & -i\omega & icl \\ ick & icl & -i\omega \end{vmatrix} = 0, \quad (12)$$

and take the determinant gives a cubic equation relating ω to k, l : $\omega[\omega^2 - c^2(k^2 + l^2) - f^2] = 0$. The three roots correspond to $\omega = 0$ and the two Poincaré roots.

The eigenvalues (in 2-d) are then

$$\omega_0 = 0 \quad \omega_+ = -\omega_- = (c^2\kappa^2 + f^2)^{1/2}$$

That is, we have a non-propagating mode and two Poincaré modes, with ω_+ propagating in the positive $\hat{\mathbf{k}}$ direction ($\mathbf{k}$ here is the wavevector, not a vertical unit vector...). The corresponding eigenvectors can be written as

$$\Psi_0 = \begin{pmatrix} -ilc \\ +ikc \\ f \end{pmatrix} \quad \Psi_+ = \begin{pmatrix} \omega_+k + ifl \\ \omega_+l - ifk \\ c\kappa^2 \end{pmatrix} \quad \Psi_- = \begin{pmatrix} \omega_-k + ifl \\ \omega_-l - ifk \\ c\kappa^2 \end{pmatrix}.$$

These eigenvectors encode the polarization relations. It is left as an exercise to confirm this for the two Poincaré roots, and that the Ψ_0 eigenvector gives that the zero root is both geostrophic and horizontally nondivergent.

It can be shown that the eigenvectors form a complete basis, implying any initial conditions can be decomposed into its geostrophic and Poincaré components. Given u, v, ϕ as functions of x, y , we first go to Fourier space to get U, V, Φ as functions of k, l . Then, for each wavevector (equivalently, each (k, l) pair), we project onto normalized versions of Ψ_0, Ψ_+, Ψ_- , giving complex coefficients A_0, A_+, A_- . We can then write

$$\begin{pmatrix} u(x, y) \\ v(x, y) \\ (1/c)\phi(x, y) \end{pmatrix} = (\Sigma_{k,l} A_0 \Psi_0 + A_+ \Psi_+ + A_- \Psi_-) e^{i(kx+ly)},$$

where you should keep in mind that the coefficients A and the vectors Ψ are functions of k, l .

Exercice 6.4, and optionally 6.5, are now within your reach!